

EXP-05

```
mysql> CREATE TABLE student (  
->     student_id INT,  
->     student_name VARCHAR(50),  
->     department VARCHAR(30),  
->     marks INT  
-> );  
Query OK, 0 rows affected (0.07 sec)  
  
mysql>  
mysql> INSERT INTO student VALUES  
-> (101, 'Arun', 'CSE', 85),  
-> (102, 'Karthik', 'ECE', 78),  
-> (103, 'Rahul', 'CSE', 92),  
-> (104, 'Vijay', 'IT', 88),  
-> (105, 'Ajay', 'CSE', 76),  
-> (106, 'Suresh', 'ECE', 82);  
Query OK, 6 rows affected (0.01 sec)  
Records: 6  Duplicates: 0  Warnings: 0  
  
mysql>  
mysql> SELECT department, COUNT(*) AS total_students  
-> FROM student  
-> GROUP BY department;  
  
+-----+-----+  
| department | total_students |  
+-----+-----+  
| CSE       | 3              |  
| ECE       | 2              |  
| IT        | 1              |  
+-----+-----+  
3 rows in set (0.01 sec)  
  
mysql>  
mysql> SELECT department, AVG(marks) AS average_marks  
-> FROM student  
-> GROUP BY department;  
  
+-----+-----+  
| department | average_marks |  
+-----+-----+  
| CSE       | 84.3333       |  
| ECE       | 80.0000       |  
| IT        | 88.0000       |  
+-----+-----+  
3 rows in set (0.00 sec)
```

```
mysql> SELECT department, AVG(marks) AS average_marks
-> FROM student
-> GROUP BY department
-> HAVING AVG(marks) > 80;
```

department	average_marks
CSE	84.3333
IT	88.0000

2 rows in set (0.00 sec)

```
mysql>
mysql> SELECT * FROM student
-> ORDER BY marks ASC;
```

student_id	student_name	department	marks
105	Ajay	CSE	76
102	Karthik	ECE	78
106	Suresh	ECE	82
101	Arun	CSE	85
104	Vijay	IT	88
103	Rahul	CSE	92

6 rows in set (0.00 sec)

```
mysql>
mysql> SELECT * FROM student
-> ORDER BY marks DESC;
```

student_id	student_name	department	marks
103	Rahul	CSE	92
104	Vijay	IT	88
101	Arun	CSE	85
106	Suresh	ECE	82
102	Karthik	ECE	78
105	Ajay	CSE	76

6 rows in set (0.00 sec)

```
mysql> SELECT department, AVG(marks) AS average_marks
-> FROM student
-> GROUP BY department
-> HAVING AVG(marks) > 80
-> ORDER BY average_marks DESC;
```

department	average_marks
IT	88.0000
CSE	84.3333

2 rows in set (0.00 sec)